

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Indicate which Mathematical Theorem *best fits* the statement.

Fermat's Theorem
Intermediate Value Theorem
Extreme Value Theorem
Mean Value Theorem

- There was a maximum temperature in the last 24 hours.
- A ball tossed in the air will be at its highest point when it stops moving for an instant.
- A fish that jumped above the surface of the water will make a splash before it is again underwater.
- If you travel 90 miles in 2 hours, at some time you were traveling 45 miles per hour.

2. Recall that Fermat's Theorem states

"If f has a local minimum or maximum value at c , and if $f'(c)$ exists, then $f'(c) = 0$."

TRUE OR FALSE: It is also true that this statement can be reverse. That is to say

"If $f'(c) = 0$, then f has a local minimum or maximum value at c ."

3. A person begins cycling from home at noon. She begins cycling quickly and travels 3 miles north. Then she makes a U-turn without stopping, and then cycles home slowly. She finishes cycling after 30 minutes. Let $g(x)$ be a function that represents the total displacement of the biker from her home versus time.
- a) Construct and label a possible graph of $g(x)$.
- b) Construct a line segment on the graph from part (a) from the starting point to the ending point. On the graph, label the point c guaranteed by the appropriate theorem.
- c) Refer to the cycling context of the problem to describe what the point c means in the problem.

4. Let $g(x) = x^2 - 4x + 3$.

a) If $a = 0$ and $b = 2$, what value c satisfies the conclusion of the MVT?

b) If $a = 0$ and $b = 3$, what value c satisfies the conclusion of the MVT?

c) If $a = 1$ and $b = 3$, what value c satisfies the conclusion of the MVT?

d) In general, what value c do you anticipate satisfies the conclusion of the MVT for any a and b with this polynomial?

e) Suppose that $p(x) = Ax^2 + Bx + C$. Show that the number c that satisfies the conclusion of the MVT for any polynomial $p(x)$ is given by $c = \frac{a+b}{2}$

5. Show that the equation $6x - \cos(x) = 0$ has exactly one real solution.

6. Suppose f is an odd function and is differentiable everywhere. Prove that for every positive number b , there exists a number c in $(-b, b)$ such that $f'(c) = \frac{f(b)}{b}$.

7. Suppose that f and g are continuous on $[a, b]$ and differentiable on (a, b) . Suppose also that $f(a) = g(a)$ and $f'(x) < g'(x)$ for $a < x < b$. Prove that $f(b) < g(b)$.